Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-4735
Project Name	Credit card Approval Prediction
Maximum Marks	2 Marks

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap 5, JavaScript	Used to build a responsive, user-friendly web interface for collecting applicant details and displaying prediction results.
2	**Backend / Server-Side**	Python, Flask	Flask handles HTTP requests, processes user input, loads the trained Random Forest model, and returns prediction results efficiently.
3	**Database / Data Storage**	Joblib (.pkl Model File)	The trained Random Forest model is stored as a `.pkl` file using Joblib, enabling fast loading and prediction without requiring a database.
4	**Cloud / Hosting / Deployment**	Render	Used to deploy the Flask web application online with minimal configuration and free cloud hosting support.

5	**Version Control & CI/CD**	GitHub	Used to manage source code, track changes, and integrate directly with Render for automatic deployment from the repository.

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn	Scikit-learn is used for training the Random Forest model, Pandas and NumPy for data preprocessing, and Matplotlib/Seaborn for data visualization and model evaluation during development.